Database Management System LAB
Spring Semester 2024-25
Lab Assignment 5

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Table Creation

Foreign Keys will be added later on after creating all tables

```
mysql> use Assignment_5
Database changed
mysql> \e;
Query OK, 0 rows affected (0.05 sec)

Query OK, 5 rows affected (0.06 sec)
Records: 5  Deleted: 0  Skipped: 0  Warnings: 0

+------+---------+-----------+------+
| Snum | Sname   | City      | Comm |
+------+---------+-----------+------+
| 1001 | Peel    | London    | 0.12 |
| 1002 | Serres  | Sanjose   | 0.13 |
| 1003 | Axelrod | Newyork   |  0.1 |
| 1004 | Motika  | Landon    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
+------+---------+-----------+------+
5 rows in set (0.06 sec)
```

```
  GNU nano 7.2
create table SalesPeople
(Snum int PRIMARY KEY,
Sname Varchar(20) UNIQUE,
City Varchar(20),
Comm float);

load data local infile 'SalesPeople.txt' into table SalesPeople;

select * from SalesPeople;
```

```
1001    Peel    London  0.12
1002    Serres  Sanjose 0.13
1004    Motika  Landon  0.11
1007    Rifkin  Barcelona       0.15
1003    Axelrod Newyork 0.10
```

```
mysql> \e;
Query OK, 0 rows affected (0.04 sec)

Query OK, 7 rows affected (0.05 sec)
Records: 7  Deleted: 0  Skipped: 0  Warnings: 0

+------+----------+---------+------+
| Cnum | Cname    | City    | Snum |
+------+----------+---------+------+
| 2001 | Hoffman  | London  | 1001 |
| 2002 | Giovanni | Rome    | 1003 |
| 2003 | Liu      | Sanjose | 1002 |
| 2004 | Grass    | Berlin  | 1002 |
| 2006 | Clemens  | London  | 1001 |
| 2007 | Pereira  | Rome    | 1004 |
| 2008 | Cisneros | Sanjose | 1007 |
+------+----------+---------+------+
7 rows in set (0.05 sec)
```

```
  GNU nano 7.2                                  /tmp/sqld7LqEA
create table Customers
(Cnum int PRIMARY KEY,
Cname Varchar(20),
City Varchar(20) NOT NULL,
Snum int);

load data local infile 'Customers.txt' into table Customers;

select * from Customers;
```

```
2001    Hoffman London  1001
2002    Giovanni        Rome    1003
2003    Liu     Sanjose 1002
2004    Grass   Berlin  1002
2006    Clemens London  1001
2008    Cisneros        Sanjose 1007
2007    Pereira Rome    1004
```

```
mysql> \e;
Query OK, 0 rows affected (0.04 sec)

Query OK, 10 rows affected (0.04 sec)
Records: 10  Deleted: 0  Skipped: 0  Warnings: 0

+------+---------+------------+------+------+
| Onum | Amt     | Odate      | Cnum | Snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3002 |  1900.1 | 1990-10-03 | 2007 | 1004 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 |    4273 | 1990-10-05 | 2006 | 1001 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3010 | 1309.95 | 1990-10-06 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-06 | 2006 | 1001 |
+------+---------+------------+------+------+
10 rows in set (0.04 sec)
```

```
  GNU nano 7.2                                  /tmp/sqlVk3RCM
create table Orders
(Onum int PRIMARY KEY,
Amt float,
Odate DATE,
Cnum int,
Snum int);

load data local infile 'Orders.txt' into table Orders;

select * from Orders;
```

```
3001    18.69   1990-10-03      2008    1007
3003    767.19  1990-10-03      2001    1001
3002    1900.10 1990-10-03      2007    1004
3005    5160.45 1990-10-03      2003    1002
3006    1098.16 1990-10-03      2008    1007
3009    1713.23 1990-10-04      2002    1003
3007    75.75   1990-10-04      2004    1002
3008    4273.00 1990-10-05      2006    1001
3010    1309.95 1990-10-06      2004    1002
3011    9891.88 1990-10-06      2006    1001
```

Adding Foreign Keys

```
  GNU nano 7.2                                    /tmp/sqlC7KXHj
alter table Customers
Add Foreign Key (Snum) References SalesPeople(Snum);

alter table Orders
Add Foreign Key (Snum) References SalesPeople(Snum);

alter table Orders
Add Foreign Key (Cnum) References Customers(Cnum);
```

```
mysql> \e;
Query OK, 7 rows affected (0.11 sec)
Records: 7  Duplicates: 0  Warnings: 0

Query OK, 10 rows affected (0.21 sec)
Records: 10  Duplicates: 0  Warnings: 0

Query OK, 10 rows affected (0.32 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Ans – 1)

```
mysql> select COUNT(*) from SalesPeople
    -> where Sname like 'A%';
+----------+
| COUNT(*) |
+----------+
|        1 |
+----------+
1 row in set (0.03 sec)
```

Ans – 2)

```
  GNU nano 7.2                                    /tm
select s.Snum,Sname from SalesPeople as s
Join Orders as o on s.Snum = o.Snum
Group by s.Snum
Having sum(Amt) > 2000;
```

```
mysql> \e;
+------+--------+
| Snum | Sname  |
+------+--------+
| 1001 | Peel   |
| 1002 | Serres |
+------+--------+
2 rows in set (0.01 sec)
```

Ans – 3)

```
  GNU nano 7.2                                    /tm
select COUNT(*) as belonging_to_newyork
from SalesPeople
where City = 'NewYork';
```

```
mysql> \e;
+----------------------+
| belonging_to_newyork |
+----------------------+
|                    1 |
+----------------------+
1 row in set (0.00 sec)
```

Ans – 4)

```
  GNU nano 7.2                                    /tmp/
select City,COUNT(*)
from SalesPeople
Group by City
Having City = 'London' or City = 'Paris';
```

```
mysql> \e;
+--------+----------+
| City   | COUNT(*) |
+--------+----------+
| London |        1 |
+--------+----------+
1 row in set (0.00 sec)
```

Ans – 5)

```
  GNU nano 7.2
select Sname,COUNT(*)
from SalesPeople as s
Join Orders as o on o.Snum = s.Snum
Group by s.Snum,Odate;
```

```
mysql> \e;
+---------+----------+
| Sname   | COUNT(*) |
+---------+----------+
| Rifkin  |        2 |
| Motika  |        1 |
| Peel    |        1 |
| Serres  |        1 |
| Serres  |        1 |
| Peel    |        1 |
| Axelrod |        1 |
| Serres  |        1 |
| Peel    |        1 |
+---------+----------+
9 rows in set (0.00 sec)
```

Ans – 6)

```
  GNU nano 7.2
select Odate as date,COUNT(*)
from Orders
Group by Odate;
```

```
mysql> \e
    -> ;
+------------+----------+
| date       | COUNT(*) |
+------------+----------+
| 1990-10-03 |        5 |
| 1990-10-04 |        2 |
| 1990-10-05 |        1 |
| 1990-10-06 |        2 |
+------------+----------+
4 rows in set (0.00 sec)
```

Ans – 7)

```
  GNU nano 7.2
select Cname
from Customers
where Cname like 'G%'
Order by Cname
Limit 1;
```

```
mysql> \e;
+----------+
| Cname    |
+----------+
| Giovanni |
+----------+
1 row in set (0.00 sec)
```

Ans – 8)

```
  GNU nano 7.2
select Snum,max(Amt)
from Orders
Group by Snum
Having Snum = 1002 or Snum = 1007;
```

```
mysql> \e;
+------+----------+
| Snum | max(Amt) |
+------+----------+
| 1007 |  1098.16 |
| 1002 |  5160.45 |
+------+----------+
2 rows in set (0.00 sec)
```

Ans – 9)

```
  GNU nano 7.2
select Snum,max(Amt)
from Orders
Group by Snum
Having max(Amt) > 3000;
```

```
mysql> \e;
+------+----------+
| Snum | max(Amt) |
+------+----------+
| 1001 |  9891.88 |
| 1002 |  5160.45 |
+------+----------+
2 rows in set (0.00 sec)
```

Ans – 10)

```
  GNU nano 7.2
select City,Comm,COUNT(*)
from SalesPeople
Group by City,Comm;
```

```
mysql> \e;
+-----------+------+----------+
| City      | Comm | COUNT(*) |
+-----------+------+----------+
| London    | 0.12 |        1 |
| Sanjose   | 0.13 |        1 |
| Newyork   |  0.1 |        1 |
| Landon    | 0.11 |        1 |
| Barcelona | 0.15 |        1 |
+-----------+------+----------+
5 rows in set (0.00 sec)
```

Ans – 11)

```
  GNU nano 7.2                                    /tmp/sq
select Sname,City
from SalesPeople
where City = Some(Select City from Customers);

select DISTINCT Sname,s.City
from SalesPeople as s
Join Customers as c on c.City = s.City;
```

```
mysql> \e;
+-------+---------+
| Sname | City    |
+-------+---------+
| Peel  | London  |
| Serres | Sanjose |
+-------+---------+
2 rows in set (0.00 sec)

+-------+---------+
| Sname | City    |
+-------+---------+
| Peel  | London  |
| Serres | Sanjose |
+-------+---------+
2 rows in set (0.00 sec)
```

Ans – 12)

```
  GNU nano 7.2                                    /
select Sname,Cname
From SalesPeople as s,Customers as c
where s.City = c.City;

select Sname,Cname
from SalesPeople as s
Join Customers as c on s.City = c.City;
```

```
mysql> \e;
+--------+----------+
| Sname  | Cname    |
+--------+----------+
| Peel   | Hoffman  |
| Serres | Liu      |
| Peel   | Clemens  |
| Serres | Cisneros |
+--------+----------+
4 rows in set (0.00 sec)

+--------+----------+
| Sname  | Cname    |
+--------+----------+
| Peel   | Hoffman  |
| Serres | Liu      |
| Peel   | Clemens  |
| Serres | Cisneros |
+--------+----------+
4 rows in set (0.00 sec)
```

Ans – 13)

```
  GNU nano 7.2                                  /tmp/sqlPi6fP9
select DISTINCT Sname
From SalesPeople as s,Customers as c
where (s.City = c.City and s.Snum = c.Snum);

select DISTINCT Sname
from SalesPeople as s
Join Customers as c on (s.City = c.City and s.Snum = c.Snum);
```

```
mysql> \e;
+--------+
| Sname  |
+--------+
| Peel   |
| Serres |
+--------+
2 rows in set (0.00 sec)

+--------+
| Sname  |
+--------+
| Peel   |
| Serres |
+--------+
2 rows in set (0.00 sec)
```

Ans – 14)

```
  GNU nano 7.2                                  /tmp/sqlPi6fP9
select DISTINCT Sname
From SalesPeople as s,Customers as c
where (s.City = c.City and s.Snum = c.Snum);

select DISTINCT Sname
from SalesPeople as s
Join Customers as c on (s.City = c.City and s.Snum = c.Snum);
```

```
mysql> \e;
+--------+
| Sname  |
+--------+
| Peel   |
| Serres |
+--------+
2 rows in set (0.00 sec)

+--------+
| Sname  |
+--------+
| Peel   |
| Serres |
+--------+
2 rows in set (0.00 sec)
```

Ans – 15) Note that there is no unassigned SalesPeople otherwise would have been NULL.

```
  GNU nano 7.2                                  /tmp/s
select Sname,Cname
From SalesPeople as s
Left Join Customers as c on c.Snum = s.Snum;
```

```
mysql> \e;
+---------+----------+
| Sname   | Cname    |
+---------+----------+
| Axelrod | Giovanni |
| Motika  | Pereira  |
| Peel    | Hoffman  |
| Peel    | Clemens  |
| Rifkin  | Cisneros |
| Serres  | Liu      |
| Serres  | Grass    |
+---------+----------+
7 rows in set (0.00 sec)
```

Ans – 16)

```
  GNU nano 7.2
select Cname
From Customers
where Snum is NULL;
```

```
mysql> \e;
Empty set (0.00 sec)
```

Ans – 17)

```
select Snum,max(Amt)
from Orders
Group by Snum;
```

```
mysql> \e;
Query OK, 0 rows affected (0.00 sec)

+------+----------+
| Snum | max(Amt) |
+------+----------+
| 1001 |  9891.88 |
| 1002 |  5160.45 |
| 1003 |  1713.23 |
| 1004 |   1900.1 |
| 1007 |  1098.16 |
+------+----------+
5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

Ans – 18)

```
  GNU nano 7.2                          /tmp/sqlSlDM9o
# Disable ONLY_FULL_GROUP_BY
SET GLOBAL sql_mode=(SELECT REPLACE(@@sql_mode,'ONLY_FULL_GROUP_BY',''));

select s.Sname,max(Amt)
From Orders as o
Left Join SalesPeople as s on o.Snum = s.Snum
Group by o.Snum;
```

```
mysql> \e;
Query OK, 0 rows affected (0.00 sec)

+---------+----------+
| Sname   | max(Amt) |
+---------+----------+
| Rifkin  |  1098.16 |
| Motika  |   1900.1 |
| Peel    |  9891.88 |
| Serres  |  5160.45 |
| Axelrod |  1713.23 |
+---------+----------+
5 rows in set (0.00 sec)
```

Ans – 19)

```
select s.Snum,Sname,Amt
From Orders as o
Left Join SalesPeople as s on o.Snum = s.Snum
where Amt >
(Select avg(Amt) from Orders
where Snum = s.Snum);
```

```
mysql> \e;
Query OK, 0 rows affected (0.00 sec)

+------+--------+---------+
| Snum | Sname  | Amt     |
+------+--------+---------+
| 1002 | Serres | 5160.45 |
| 1007 | Rifkin | 1098.16 |
| 1001 | Peel   | 9891.88 |
+------+--------+---------+
3 rows in set (0.00 sec)
```

Ans – 20)

```
  GNU nano 7.2                          /tmp/sqljHKs34
# Disable ONLY_FULL_GROUP_BY
SET GLOBAL sql_mode=(SELECT REPLACE(@@sql_mode,'ONLY_FULL_GROUP_BY',''));

select Sname
From SalesPeople as s
Left Join Customers as c on c.Snum = s.Snum
Group by s.Snum
Having Count(*) > 2;
```

```
mysql> \e;
Query OK, 0 rows affected (0.00 sec)

Empty set (0.00 sec)
```

Ans – 1)
Ans – 1)
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